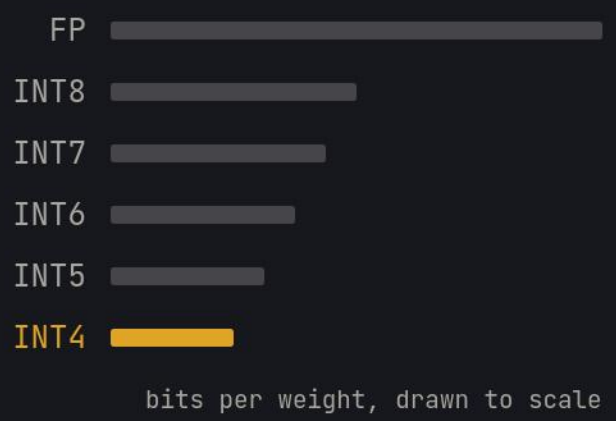
ANLP viva

Language Features under Weight Quantization

Activation Agreement and Intervention Effects

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The question

When a model's weights are rounded to fewer bits, do the internal features that explain it still do the same job?

27 languages · Gemma-2-2B (main) · Gemma-2-9B check · fixed sparse autoencoder

Related work

Prior work gave the units and the question; we test whether they still do the same thing

Marchisio et al. 2024

Quantization affects languages unequally

`asks what changes`

Deng et al. 2025

Found language-specific SAE features and showed that editing them changes output

`gives us the units`

Duan 2026

Measured activation survival of features under quantization

`asks if features fire the same`

Our contribution

Functional preservation, not just activation: does a feature still do the same thing?

`asks if features do the same`

Methodological grounding

Makelov et al. 2025
Judge by edits

Leask et al. 2025
SAE features are not canonical

Karvonen et al. 2025
Proxy scores ≠ useful features

Research questions

Four questions, from activation to effect

RQ1

Which language features retain their activation patterns after quantization?

Measured by activation correlation R_1

RQ2

How does editing language features affect text prediction and comprehension?

Measured by excess harm, dose-response and Belebele accuracy

RQ3

How well do activation measures predict whether a feature's output effect is preserved?

Measured by effect cosine W_1, swap control, patching

RQ4

How much does the same feature edit change when the model is quantized?

Measured by context, later-weight and interaction terms

Setup

Three models, 27 languages, three quantizers

Models

Gemma-2-2B
main · layer-12 JumpReLU SAE
16,384 features

Gemma-2-9B
confirmation

Llama-3.2-1B
cross-check

Data

27 FLORES-200 languages
9 per resource tier

300 sentences
per language

Belebele
18 languages

Quantizers

Legacy RTN
reproduces Duan

Matched RTN
one scale per output channel

GPTQ
calibrated

Methodology

Excess harm: the loss from removing a language's features, net of a control

1
Pick 64 features
The ones that fire most for one language

2
Zero-ablate
Turn them off

3
Measure NLL rise
Next-word prediction gets worse

4
Subtract control harm
Same edit on 64 ordinary features

$$E_\ell = \Delta\mathrm{NLL}_\ell(\mathrm{group}_\ell) - \mathrm{mean}\Big(\Delta\mathrm{NLL}_\ell(\mathrm{controls}_\ell)\Big)$$

E_ℓ is the excess harm for language ℓ in nats per token; positive means the language edit hurts more than the control edit

Repeat at FP and INT8 through INT4

FP INT8 INT7 INT6 INT5 INT4

A correction to earlier work

One missing token (BOS) made the baseline look 28× worse

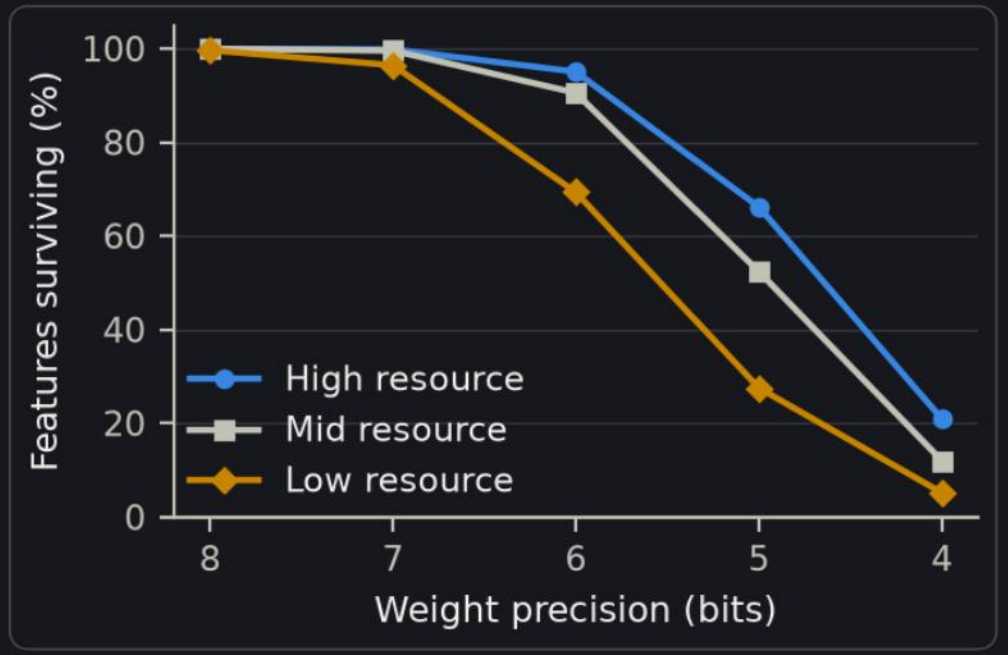
	No BOS	With BOS
FP perplexity	458.4	16.1
INT6 survival	51%	97.5%
INT7 survival	81%	>99.9%

- Duan 2026 omitted the beginning-of-sequence token
- We reproduced their code and matched their numbers
- Then we traced the discrepancy to the missing BOS

All our results use the corrected inputs

RQ1 Activation survival

Language-selective features survive almost as well as controls



Features surviving ($R_1 > 0.9$) by resource tier

-0.28 points at INT7

-1.44 points at INT6

Selective minus control survival, not significant after correction

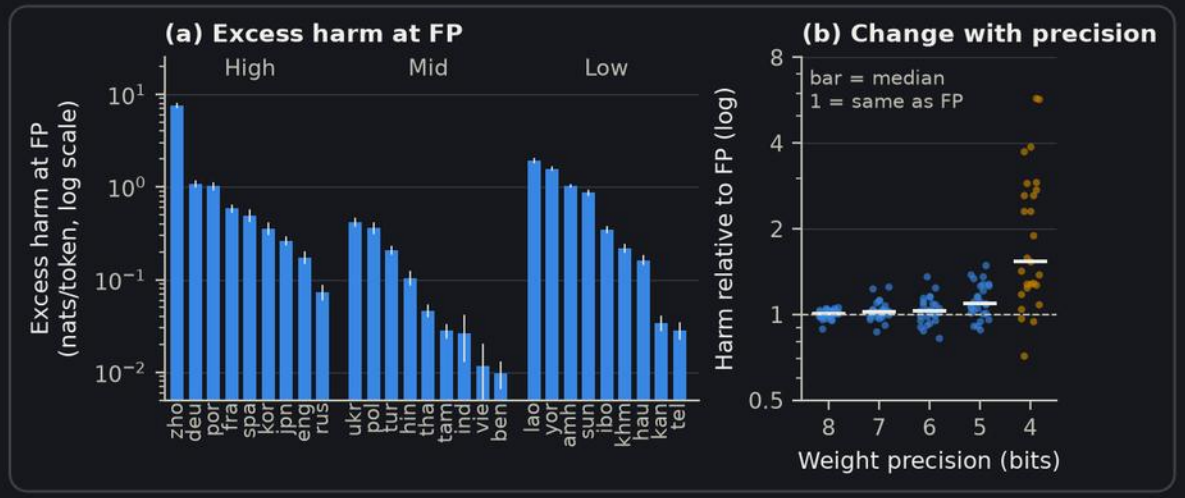
$$R_1 = \mathrm{corr}_t\left(a_f^{\mathrm{FP}}(t),\, a_f^{b}(t)\right)$$

a feature survives if $R_1 > 0.9$

- The gap tracks activation strength, not language selectivity
- Weaker features survive less: Spearman $\rho = 0.35$ at INT6

RQ2 Prediction

Removing language features hurts next-word prediction in all 27 languages



(a) FP excess harm per language · (b) harm relative to FP, one dot per language

27/27 languages with positive FP excess harm

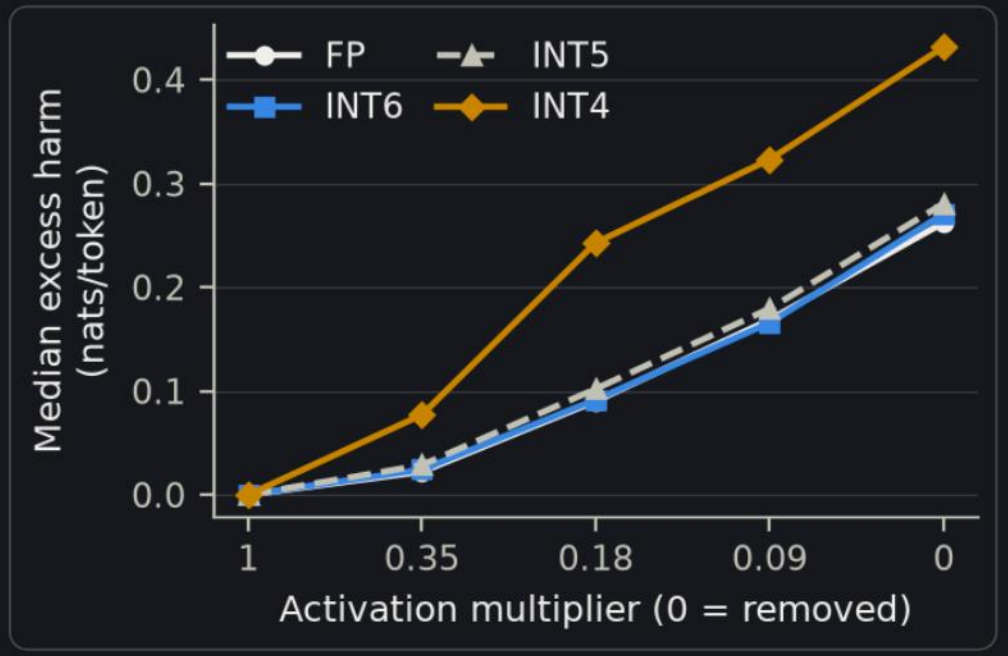
0.704 mean excess harm, nats/token [0.684, 0.723]

1.54 INT4 median ratio to FP [1.43, 1.62]

≈FP through INT5, then growth at INT4

RQ2 Dose-response

More suppression, more harm, in every language



Median excess harm against activation multiplier

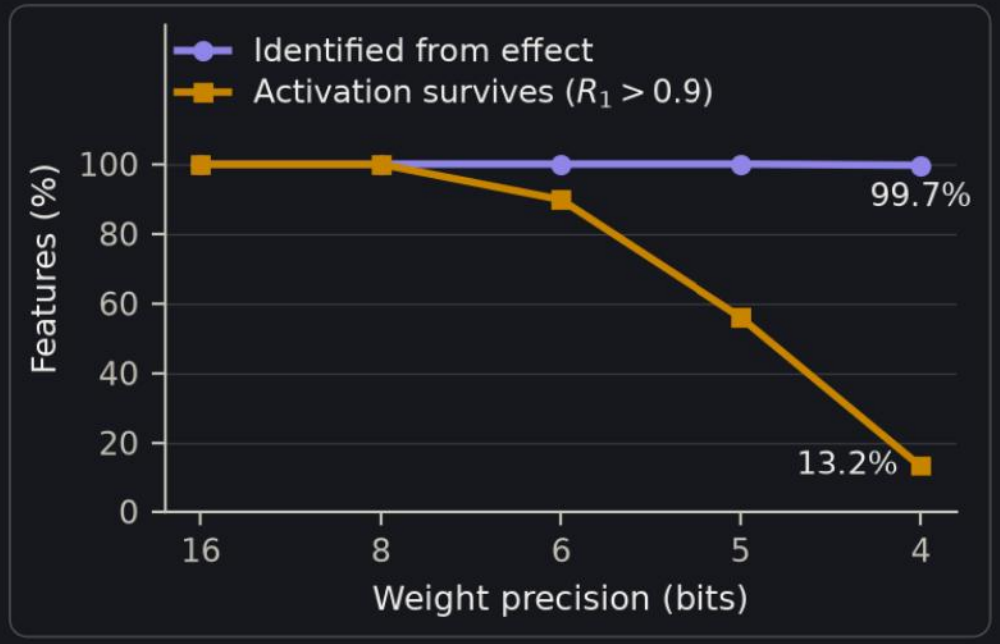
Activation multiplier	Median harm
0.354	0.022
0.177	0.090
0.088	0.169
0	0.262

FP, nats/token. Holds in **all 27 languages**; at INT4 the same step hurts more

A graded response is evidence of a real causal role

RQ3 Headline result

Whether a feature fires the same and whether it works the same are different questions



1,487-feature sample at INT4

13.2%

pass the activation cutoff, $R_1 > 0.9$

99.7%

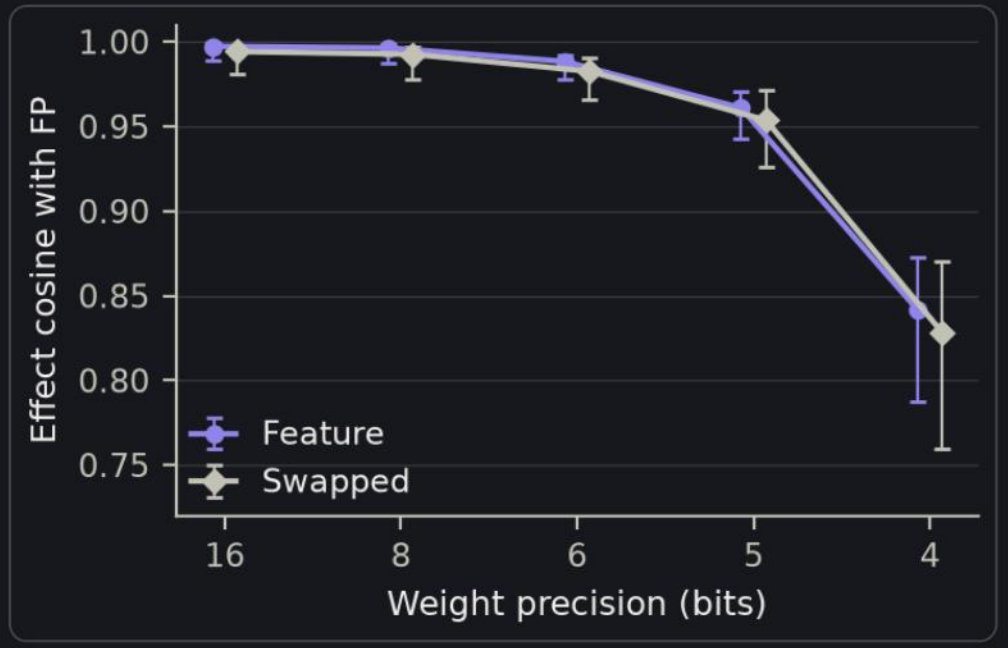
identified by output effect

$$W_1(f, b) = \cos\left(\mathbf{u}_f^{\mathrm{FP}}, \mathbf{u}_f^{b}\right)$$

cosine between FP and quantized effect fingerprints $\mathbf{u}$

RQ3 Direction-swap control

The 4-bit drift is generic, not specific to the features



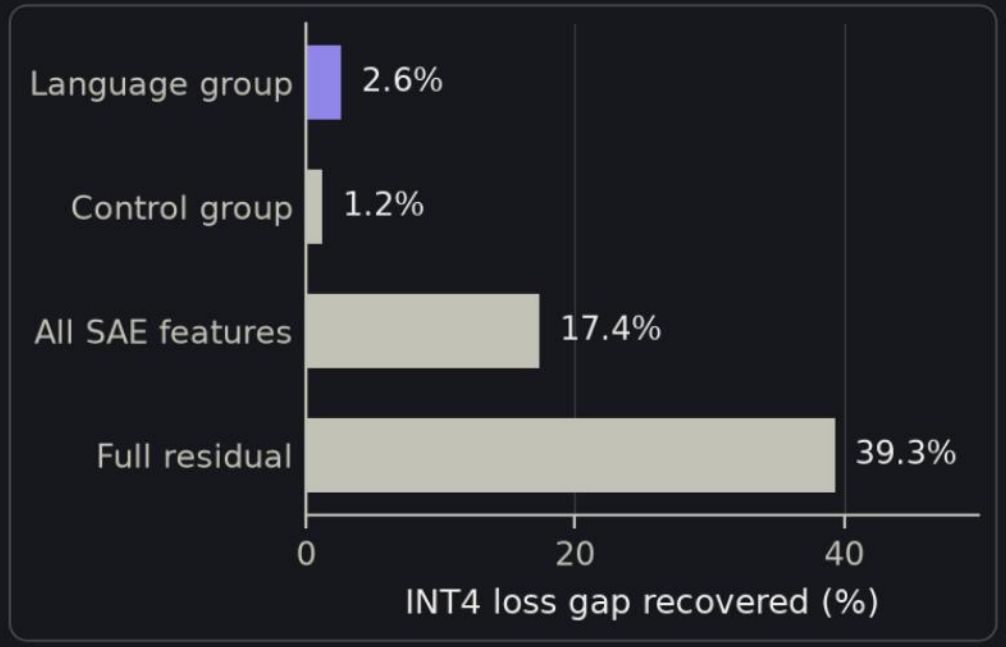
Median effect cosine with FP (IQR), 404 features; swapped direction at the same positions and strengths

0.842	feature cosine
0.828	swapped-control cosine
0.014	gap at INT4 0.003 to 0.014 at every precision

The INT4 drift comes from downstream computation, not from the features

RQ3 Activation patching

Fixing the features fixes little of the INT4 loss



Share of the INT4 loss gap recovered by restoring FP activations

2.6%

recovered by restoring the language group
[2.4, 2.8]

39.3%

recovered by restoring the full residual
[38.5, 40.0]

The features survive functionally but carry almost none of the compression damage

RQ4 Where the change comes from

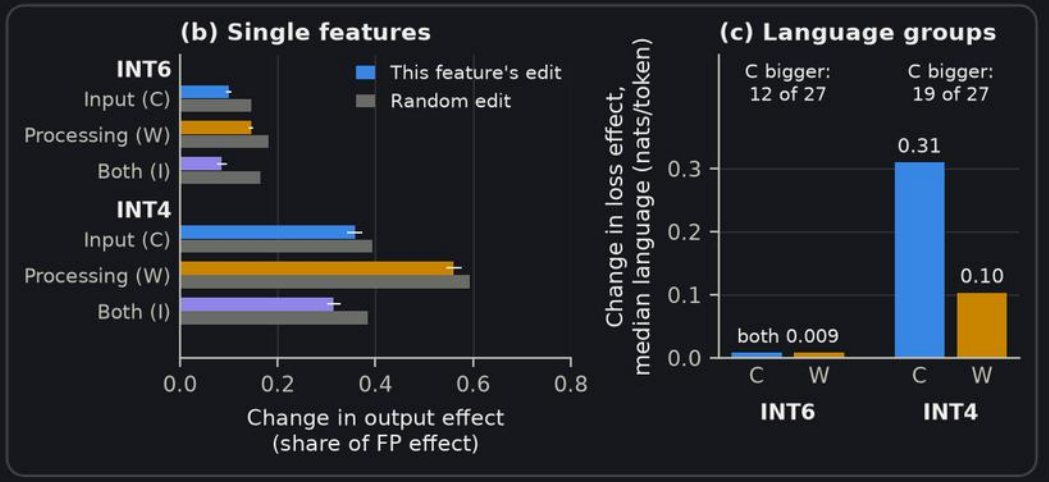
Group loss changes through context; single features through later weights

Same edit, same strength. $D[p, s]$ is the output change with earlier layers p and later weights s, each FP or Q

	later weights FP	later weights Q
context FP	D[FP,FP] reference	D[FP,Q] gives W
context Q	D[Q,FP] gives C	D[Q,Q] adds I

$$C = D[\mathrm{Q},\mathrm{FP}] - D[\mathrm{FP},\mathrm{FP}]$$
$$W = D[\mathrm{FP},\mathrm{Q}] - D[\mathrm{FP},\mathrm{FP}]$$

C context, W later weights, I their interaction



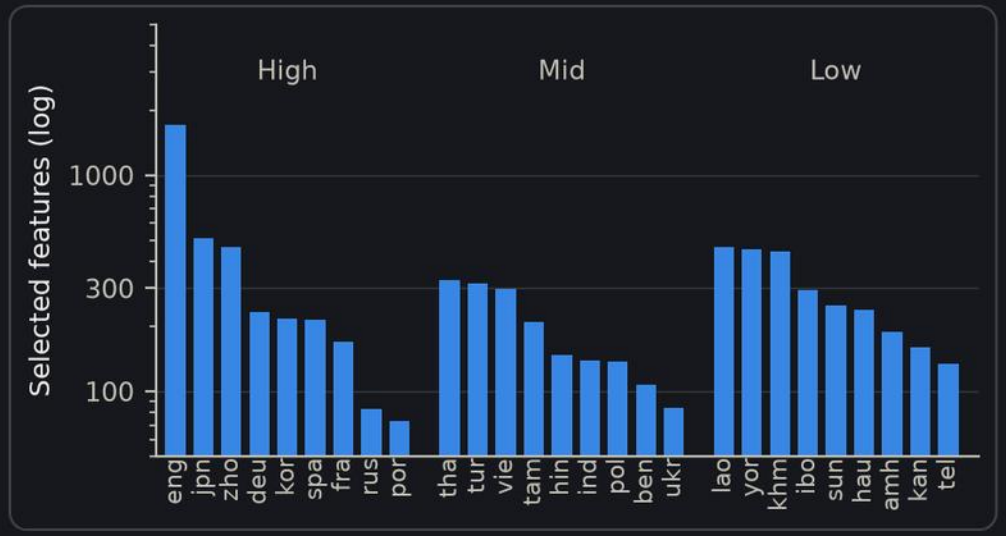
Single features (b) and language groups (c), legacy RTN

Group loss
context term larger at INT4

Single features
later-weight term larger

Descriptive

Feature counts do not follow resources



Selected language-specific features per language, log scale

Lao 464 · Khmer 444 · Yoruba 453
more features than German or Spanish

Portuguese 73 · Russian 83
close relatives share features, so neither owns them

$\rho = -0.07$
Spearman, feature count against resource tier

Count reflects how distinct a language looks to the model, not its resources

Belebele reading comprehension

The only pre-registered test came back null

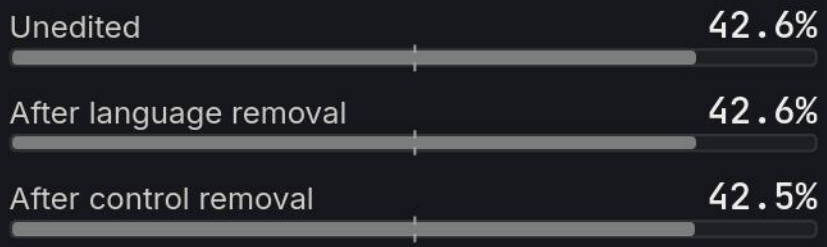
Pre-registered · confirmatory · null result

FP excess correct-option loss

-0.0001 nats/item [-0.0035, +0.0031]



Accuracy over 18 languages, 4-way choice



axis 0 to 50%, dashed line is chance (25%)

- The required positive FP effect is absent
- So the planned conditional statements cannot be interpreted
- Larger edits (1,024+ features) do lower accuracy: the test has sensitivity, just not for K64

The features matter for next-word prediction, not clearly for this task

Cross-model validation

The pattern holds on Gemma-2-9B and across quantizers, but not on Llama

Gemma-2-9B

27/27 positive

- INT6 near FP
- INT4 ratio 1.586 [1.238, 2.264]

All frozen predictions met

Three quantizers

3/3 same pattern

- Legacy, Matched and GPTQ
- All give the same qualitative pattern

Consistent

Llama-3.2-1B

21/27 rule: 24

- INT4 ratio 0.86
- Model, SAE and tokenizer all differ

Verdict: not consistent

Summary

Every INT4 measure improves from Legacy to Matched to GPTQ

INT4 measurement	Legacy	Matched	GPTQ
Active-feature survival (%)	12.7	18.6	30.5
Mean loss increase (nats/tok)	0.590	0.385	0.130
Median group-effect ratio	1.54	1.36	1.22
Median effect cosine (W_1)	0.839	0.875	0.923
Transfer cosine	0.768	0.822	0.877
Belebele accuracy (%)	23.6	29.1	31.8

Gemma-2-2B; Belebele uses 18 languages; the rows answer different questions

Limitations

What these results do not show

01	One layer, one SAE	Language-specific units concentrate in early and late layers; we tested a middle one
02	Fixed-length inputs	Truncation is controlled across precisions but untested with complete sentences
03	Mostly exploratory	Most FLORES analyses are exploratory; only Belebele was confirmatory
04	Bounded to Gemma	The Llama inconsistency limits the finding to the Gemma family
05	Simulated quantization	Weights are rounded in floating point, not run with integer kernels

Claims

What we claim

1. Language features help next-word prediction in **all 27 languages**, on both Gemma sizes
2. Dose-response confirmed in **every language**
3. Activation survival understates preservation: 13% “survive”, 99.7% still work
4. The 4-bit drift is generic (direction-swap control), and restoring features repairs only 2.6%
5. Activation agreement, task performance and effect need **separate measurements**

Reproducibility

Reported with its failures, from frozen predictions

- ✓ Reproduced Duan 2026 before reporting its BOS problem
- ✓ Predictions and code hashes frozen before data existed
- ✓ Two faulty 9B predictions documented, not counted
- ✓ Null and failed results reported alongside positive ones

Code
`github.com/TH4KUR/always-late`

Artifacts
`huggingface.co/datasets/karma-skz/always-late-artifacts`

Runs
`wandb.ai/karma-skz-/anlp-project`

Questions